PROJECT IMPLEMENTATION STATUS

Suicide-Risk Detection Using NLP-Based Text and Speech Analysis

Detailed Project Steps, Current Progress, Completed Work, Model Results and Remaining Roadmap

Presenting Team

23761A1251 - S. Thanmayi
24765A1201 - Ch. Venu Madhav
23761A1235 - M. Samuel Babu

Guide
Dr. G. Rajendra, Associate Professor

Current milestone: Dataset and preprocessing phase complete; text-model implementation and comparison is actively in progress.

Prepared as the working implementation roadmap for the complete project.

1. Executive Project Status

The project architecture is being implemented exactly as a multi-signal NLP and speech screening system. Text and speech are processed through separate branches, and their outputs are combined only at the final decision layer.

Area	Status	Current position
Project architecture	**COMPLETE** Review-1 architecture fixed and being followed.	Text branch + Speech branch + Fusion + Risk output
Dataset audit	**COMPLETE** Text and EATD datasets fully audited.	Leakage, missing files and audio anomalies checked
Data preprocessing	**COMPLETE** Text cleaning and EATD feature extraction done.	Model-ready files created
Text-model implementation	**IN PROGRESS** BiLSTM, SVM and BERT-embedding baselines tested.	Best current text model: calibrated Linear SVM
Speech-model implementation	**NEXT** Feature data ready; classifier training still to start.	Participant-level EATD features prepared
Fusion / final decision	**PENDING** Will start after both branch models are locked.	Late-fusion design already defined
UI / demo / final report	**PENDING** After model completion.	Final integration stage

Approximate overall implementation progress: about 45-50%. Dataset work is complete; model-building, fusion, UI and final validation remain.

2. Final Project Architecture

Project title: Suicide-Risk Detection Using NLP-Based Text and Speech Analysis

The system supports three operating modes: text-only, speech-only and text + speech.

Branch	Input	Model purpose	Output
Text NLP	Typed / submitted text	Identify suicide-related language patterns	Suicide-text probability P_text
Speech	Recorded / uploaded voice	Identify depression/distress acoustic patterns	Distress probability P_speech
Fusion	P_text + P_speech	Combine separate evidence signals	Low / Moderate / High screening level

Important scientific rule: SDCNL and EATD are not merged row-by-row because they do not share the same participants or ground-truth target. The project uses late decision fusion.

3. Datasets Used

Dataset A - SDCNL / Reddit text dataset

Purpose: primary suicide-related text classification dataset. The direct binary target is **is_suicide**.

File	Rows	Class 0	Class 1	Use
training-set-clean.csv	1513	727	786	Model training
testing-set-clean-leakage-free.csv	379	186	193	Held-out final test
combined-set-clean.csv	1892	913	979	EDA/reference only

Split logic: the original text split is already about 80:20. It is preserved. No additional random 80:20 split is needed for the final test set.

Dataset B - EATD-Corpus

Purpose: complementary speech + transcript depression/distress branch. It contains Mandarin speech, transcripts and SDS scores; it is not a direct suicide-labelled dataset.

Split	Folders	Usable	Depressed	Non-depressed
Train	83	80	19	61
Validation	79	79	11	68

4. Preprocessing and Audit Completed

Text preprocessing complete

Unwanted index columns were removed. model_text was derived from megatext_clean. Empty inputs and duplicate model_text + label rows were handled. Labels remained unchanged. Train-test leakage check confirmed zero normalized model_text overlap.

Speech preprocessing complete

For EATD, the project prefers dataset-provided *_out.wav files, falls back to raw *.wav only if needed, loads model audio as mono 16 kHz, applies peak normalization and extracts participant-level features.

Speech features extracted	Details
MFCC	13 means + 13 standard deviations
Pitch	Mean and standard deviation
RMS energy	Mean and standard deviation
Zero-crossing rate	Mean and standard deviation
Spectral centroid	Mean and standard deviation
Duration	Audio duration in seconds

EATD anomalies handled: train folders 1, 10 and 101 are empty and excluded; participant 102 lacks raw label.txt but has new_label.txt; validation participant 79 has a zero-duration positive_out.wav and uses valid positive.wav as fallback.

5. Text Model Implementation - Current Results

The text branch has already moved beyond preprocessing. Multiple models were implemented on the same cleaned train set and evaluated on the same held-out leakage-free test set.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Status
BiLSTM (corrected packed sequence)	59.37%	57.77%	75.13%	65.32%	63.58%	Comparison model
TF-IDF bigram + calibrated Linear SVM	72.56%	71.92%	75.65%	73.74%	78.22%	**Best current model**
BERT frozen mean-pooled embeddings + Logistic Regression	64.12%	63.77%	68.39%	66.00%	72.27%	Transformer baseline

Current conclusion: the calibrated Linear SVM is the strongest genuine text model so far. The BERT result above is a frozen-embedding baseline, not a fully fine-tuned BERT classifier. Therefore the text branch is not yet locked.

Why BiLSTM is lower: the available text dataset is relatively small for learning strong embeddings from scratch. The corrected BiLSTM is retained because it matches the proposed deep-learning architecture and provides a valid comparison.

6. Detailed Stage-by-Stage Progress

#	Stage	Status	Progress	Details
1	Problem definition and Review-1 architecture	COMPLETE	100%	Title, abstract, existing/proposed system, literature, system design and implementation flow fixed.
2	Dataset selection	COMPLETE	100%	SDCNL for text-suicide signals; EATD for speech distress signals.
3	Dataset audit and cleaning	COMPLETE	100%	Rows, labels, missing values, duplicates, leakage and EATD folder integrity checked.
4	Text preprocessing	COMPLETE	100%	Final leakage-free train/test files created.
5	Speech preprocessing / feature extraction	COMPLETE	100%	EATD participant-level features created.
6	Text model development	IN PROGRESS	75%	BiLSTM, SVM and BERT embedding baselines evaluated; model finalization pending.
7	Speech model development	NEXT	10%	Feature data is ready; model training and validation remain.
8	Fusion / decision layer	PENDING	0%	Will combine text and speech probabilities after both branches are locked.
9	Application / UI	PENDING	0%	Text, voice and combined input interface with risk display.
10	Final testing and robustness	PENDING	0%	Invalid audio, empty input, long text, model loading and end-to-end tests.

#	Stage	Status	Progress	Details
11	Final metrics / report / PPT / demo	**PENDING**	20%	Implementation document and Review material exist; final results still pending.

7. Next Steps to Complete the Project

Step A - Finalize the text branch

1. Run a proper fine-tuned lightweight transformer or a stronger pretrained NLP classifier if CPU resources allow.
2. Keep the held-out test set untouched during tuning.
3. Compare BiLSTM, SVM and transformer results on the same metrics.
4. Lock the best text model and save all inference artifacts.

Step B - Train the speech branch

1. Use EATD participant-level train features.
2. Standardize features using a scaler fit only on train data.
3. Train candidate models such as logistic regression, SVM, random forest / gradient boosting and a compact dense neural network.
4. Evaluate only on official EATD validation participants.
5. Select the best calibrated speech distress model.

Step C - Build the fusion layer

Text branch returns P_text. Speech branch returns P_speech. When both inputs are available, a late-fusion rule combines them. Fusion weights and Low / Moderate / High boundaries will be calibrated and documented, not invented as clinical cutoffs.

Step D - Build the end-user application

The UI will support text-only, speech-only and text + speech modes. It will show branch scores, combined screening level, a concise support message and a research/non-diagnostic notice.

Step E - Final validation and presentation

Generate final confusion matrices, metric tables, model comparison charts, end-to-end demo cases, project report and final PPT. The final presentation must use only verified results produced by the implementation.

8. Final Deliverables Expected

Deliverable	What it will contain
Source code	Text preprocessing, text model, speech model, fusion, inference and UI code.
Trained text model	Best verified classifier plus vectorizer/tokenizer/encoder assets.
Trained speech model	Best EATD distress classifier plus scaler and feature schema.
Fusion module	Late-fusion logic for text + speech mode.
Web/demo application	Text box, audio upload/recording and final screening result.
Evaluation package	Accuracy, Precision, Recall, F1, ROC-AUC, confusion matrices and comparison charts.
Final report and PPT	Architecture, methodology, verified results, limitations and demo flow.

9. Current Project Files and Working Structure

Current working root on the connected system:

C:\Users\RAMESH\Desktop\Projects\Thanmayi 8000

Important current locations:

Path	Purpose
Datasets\	Raw and cleaned SDCNL data, EATD train/validation and preprocessing reports.
Datasets\Processed_EATD\	Audio-level and participant-level EATD feature files.
src\	Text training scripts and next implementation code.
models\text_model\	BiLSTM, SVM, BERT embedding model artifacts.
reports\text_model\	Metrics, classification reports, predictions and plots.

10. Rules We Will Follow Until Project Completion

1. No data leakage: held-out test data is never used for training or tuning.
2. No unnecessary SMOTE on the already balanced text data.
3. Preserve the official EATD train/validation separation.
4. Do not claim EATD has suicide labels; it provides depression/distress evidence.
5. Do not directly merge SDCNL and EATD rows.
6. Report only metrics actually produced by model evaluation.
7. Keep reproducible preprocessing, saved artifacts and model configurations.
8. Treat final output as a research screening prototype, not a clinical diagnosis.

Immediate next implementation action: finalize the text branch, then train and validate the EATD speech branch.

CURRENT STATUS SUMMARY

Datasets: COMPLETE
Preprocessing: COMPLETE
Text model: IN PROGRESS - current best SVM accuracy 72.56%, ROC-AUC 78.22%
Speech model: NEXT
Fusion / UI / final testing: PENDING

This document reflects the verified implementation state at the time of preparation.